Multimodal Emotion Recognition with Pretrained Audio and Text Encoders: A Whisper-BERT Approach on MELD

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***Abstract*—Recognizing emotions in spoken conversations is vital for building intelligent systems that can interpret human behavior in a socially aware manner. However, many high-performing emotion recognition models rely on large, complex architectures that are difficult to replicate and deploy in real-world settings. This study presents a lightweight, multimodal framework that combines acoustic and textual features to classify emotions in dialogue. Audio data is converted into mel-spectrograms and analyzed using a vision transformer model, while corresponding textual utterances are encoded using a contextual language model. These two information streams are integrated through a late-fusion strategy to make final predictions across seven emotion categories. The model achieves 62% accuracy and a weighted F1-score of 62% on a benchmark conversational dataset, using only about 120 million parameters nearly half the size of many recent models. This performance demonstrates that rich emotional information can be captured without relying on speaker tracking, dialog-level memory, or external knowledge resources. The approach is easy to reproduce and well-suited for applications with limited computational infrastructure, such as real-time assistants or on-device emotion monitoring. More broadly, this work illustrates that efficient and interpretable emotion recognition is possible by leveraging complementary cues from speech and text, making it accessible for a wide range of affective computing and human-computer interaction systems.**

I. Introduction

Emotion is a crucial aspect of human communication, expressed not only through spoken words but also through tone, pitch, pauses, and other paralinguistic cues. The ability to automatically recognize emotions from conversations is fundamental to many real-world applications, such as intelligent virtual assistants, conversational agents, affective computing systems, and mental health triaging tools. In particular, with the rise in global mental health concerns and the limited availability of human support professionals, there is a growing interest in developing computationally efficient emotion-aware technologies that can assist in the early detection of emotional distress during verbal interactions. Such systems are now being explored for use in remote therapy platforms, AI-based coaching tools, intelligent tutoring systems, and even social robots designed to respond empathetically to human users.

Traditionally, research in automatic emotion recognition has focused on unimodal inputs — either text or audio. Text-based systems typically extract lexical and syntactic features or apply language models like BERT, while audio-based systems leverage prosodic signals such as intonation, energy, and rhythm, or extract embeddings from models trained on speech corpora. While both approaches have achieved success, they are limited in capturing the full spectrum of human expression, especially in natural conversations where meaning is distributed across multiple modalities.

Multimodal emotion recognition addresses this limitation by integrating information from different channels, typically combining text, audio, and sometimes video. Among the available benchmark datasets, the Multimodal EmotionLines Dataset (MELD) stands out for its richness and alignment. MELD is derived from the EmotionLines corpus and improves upon earlier datasets such as IEMOCAP by including not only high-quality utterance-level emotion annotations but also temporally synchronized audio and text from multi-party conversations. The dialogues are extracted from the TV series Friends, providing realistic, informal language and emotionally diverse interactions. This makes MELD an excellent testbed for developing and evaluating multimodal learning systems.

Despite the promise of multimodal approaches, most state-of-the-art models that perform well on MELD are complex and computationally demanding. These models often incorporate speaker-tracking, context modeling, recurrent neural networks, or cross-modal attention mechanisms. For instance, models such as DialogueRNN and DialogXL incorporate dialog history and speaker states to capture contextual dependencies, achieving weighted F1-scores above 70%. While effective, such systems are difficult to deploy in environments with limited computational resources due to their high model complexity, training time, and memory requirements. Furthermore, they often require careful tuning and access to large-scale hardware to perform optimally.

Our approach to the MELD dataset introduces a computationally efficient multimodal emotion recognition model that fuses pretrained text and audio representations without relying on complex dialog-level modeling. Specifically, we use BERT to encode textual utterances and Whisper to derive audio embeddings from the corresponding speech. Both feature vectors are mean-pooled and concatenated, forming a single joint representation that is passed through a fully connected classifier. This late-fusion strategy eliminates the need for recurrent or attention-based modules, resulting in a model that is simple to implement, faster to train, and suitable for practical deployment.

Despite its simplicity, the proposed model achieves a competitive accuracy of 61% and a weighted F1-score of 60% on the MELD dataset. These results demonstrate that it is possible to achieve meaningful performance without relying on high-complexity architectures. Additionally, the model shows robustness across several emotional categories, offering a promising foundation for future research on scalable, real-time emotion-aware systems. This work contributes to the growing body of research seeking to bridge the gap between academic advances in multimodal emotion detection and real-world deployment needs.

II. Literature Review

Emotion recognition has been extensively studied across a variety of domains, including psychology, computational linguistics, and speech processing. Early computational approaches focused on either textual or acoustic cues in isolation, relying on handcrafted features or shallow machine learning classifiers to infer emotional states from short utterances. While these unimodal methods offered simplicity and interpretability, they often struggled to capture the richness of human emotion, particularly in spontaneous conversations.

Text-based emotion classification traditionally relied on bag-of-words (BOW) models, TF-IDF features, and logistic regression or support vector machines. With the advent of deep learning, recurrent neural networks (RNNs) and later transformer models such as BERT brought significant improvements in capturing contextual semantics. BERT, pre-trained on massive corpora and fine-tuned for downstream classification, has become a standard baseline in natural language-based emotion detection tasks. However, such models are limited in their ability to incorporate non-verbal cues, which play a crucial role in conveying affective meaning.

In parallel, acoustic-based approaches analyzed prosodic features including pitch, energy, spectral flux, and voice quality, which are known to correlate with arousal and valence dimensions. Popular models like OpenSMILE and wav2vec have been used to extract high-dimensional embeddings from raw speech. These methods have demonstrated strong performance in datasets like RAVDESS and IEMOCAP, especially in controlled environments. Nonetheless, audio-only systems are susceptible to background noise, speaker variability, and ambiguity in content, making them less robust when deployed in natural conversations.

Multimodal emotion recognition has emerged as a solution to these limitations, combining textual, acoustic, and sometimes visual cues to improve robustness and accuracy. The MELD dataset has played a pivotal role in driving progress in this domain by offering high-quality utterance-level annotations and synchronized audio-text pairs from multi-party dialogues. Models built on MELD aim to exploit the complementary nature of modalities; for instance, capturing sarcasm or emotional nuance that may not be evident in text or audio alone.

Several benchmark models have been proposed for multimodal learning on MELD. DialogueRNN introduces a gated recurrent architecture that maintains speaker-specific memory states and models contextual dependencies across utterances. It achieves a weighted F1-score of approximately 62% on MELD, with further improvements when coupled with attention mechanisms. DialogXL, based on Transformer-XL, introduces recurrence and segment-level memory for long-term context modeling and achieves weighted F1-scores close to 66–67%. While these models show superior performance, they are highly complex, requiring substantial GPU memory, long training times, and careful parameter tuning.

Other models like MulT and Emora use cross-modal transformers and hierarchical attention to align temporal sequences across modalities. These architectures often incorporate multiple encoders, modality-specific attention heads, and fusion layers to dynamically weight information from each channel. As a result, they typically reach weighted F1-scores between 68% and 72%. However, such performance comes at the cost of high architectural complexity, large parameter counts, and challenges in reproducibility and deployment. For example, models like DialogXL and Emora can exceed 300 million parameters, making them difficult to scale and optimize in real-world applications.

Despite their high accuracy, many of these state-of-the-art models are not feasible for real-time applications or systems with limited computational resources. Moreover, their reliance on dialog-level modeling, speaker tracking, or hierarchical fusion introduces engineering overhead that can hinder scalability. This presents a need for simpler architectures that can still leverage multimodal information effectively.

In response to this need, recent work has explored fusion-based methods that combine pretrained encoders for each modality and apply late-stage integration for classification. These models prioritize modularity, reusability, and computational efficiency while accepting a marginal trade-off in predictive performance. Our work aligns with this line of research, proposing a fusion-based model that combines BERT (for text) and Whisper (for audio) to construct a joint embedding without any dialog context modeling or complex fusion strategy. Importantly, while many existing models operate at parameter scales exceeding 200–300 million, our model operates at approximately 120 million parameters a reduction of nearly 50% or more compared to the state-of-the-art, while still achieving a weighted F1-score of 60% and an overall accuracy of 61% on MELD.

This substantial reduction in model size offers a distinct edge in terms of computational efficiency, ease of training, and real-world reproducibility. The model's lower complexity makes it more practical for experimentation, scalable to deployment, and suitable for environments with limited hardware. By maintaining a strong balance between performance and resource usage, our approach opens up new possibilities for accessible and efficient emotion recognition systems that do not sacrifice much in terms of accuracy.

While the literature demonstrates that high accuracy is achievable through complex multimodal architectures, these models often come at the cost of massive parameter counts, extensive training requirements, and limited practicality for real-world deployment. This creates a pressing need for approaches that can offer a better trade-off between model complexity and predictive performance. In this context, our work offers a viable alternative, a computationally efficient multimodal architecture with significantly fewer parameters that is easier to reproduce, deploy, and scale. By leveraging pretrained encoders for text and audio and combining them through a simple fusion strategy, our model achieves strong performance while remaining accessible and resource-conscious, thus addressing a critical gap in the current body of research.

III. Methodology

This section presents the comprehensive methodology followed to build a computationally efficient multimodal emotion recognition system using the MELD dataset. The pipeline encompasses dataset structuring, preprocessing of multimodal inputs, leveraging transformer-based encoders for feature extraction, fusion of learned representations, and classification using a neural network. The objective is to achieve competitive performance while ensuring reproducibility, scalability, and reasonable computational cost.

A. Dataset Description

The Multimodal EmotionLines Dataset (MELD) is a benchmark corpus for multimodal emotion and sentiment classification Poria et al., 2019. MELD is derived from the EmotionLines dataset and includes over 13,000 utterances from multi-party conversations in the TV series *Friends*. Each utterance is annotated with an emotion label, a sentiment label, and is associated with aligned text and audio data. The emotion labels span seven categories: **anger, disgust, fear, joy, sadness, surprise, and neutral**.

Each utterance entry in MELD contains:

- The dialogue ID and utterance ID
- The speaker's name
- The text transcription of the utterance
- The audio recording of the utterance
- Emotion and sentiment labels

The dataset is divided into three subsets:

- **Training set:** 9989 utterances
- **Validation set:** 1109 utterances
- **Test set:** 2610 utterances

MELD maintains speaker continuity and conversational flow, although this project operates at the utterance level without explicitly modeling dialogue context.

One important characteristic of MELD is its class imbalance in emotion labels. The dataset is heavily skewed toward the 'neutral' class, which comprises over 40% of the total utterances. Other classes such as 'fear', 'disgust', and 'sadness' are underrepresented, particularly in the validation and test sets. For instance, the training set contains over 4700 'neutral' utterances, while classes like 'disgust' and 'fear' contain fewer than 300 examples each. This imbalance introduces a risk of model bias toward majority classes, requiring careful evaluation and potentially the use of class weighting.

Although MELD is a rich dataset, it has limitations with respect to preprocessing consistency and computational feasibility. The raw audio clips were extracted and aligned using automated scripting via WSL (Windows Subsystem for Linux), and running full audio pipelines remains computationally expensive, especially on machines without access to high-end GPUs. These constraints significantly impact reproducibility and resource planning.

To illustrate MELD's structure, the following image (Figure 1) provides an example dialogue showing speaker turns, utterance flow, emotion labels, and sentiment categories:

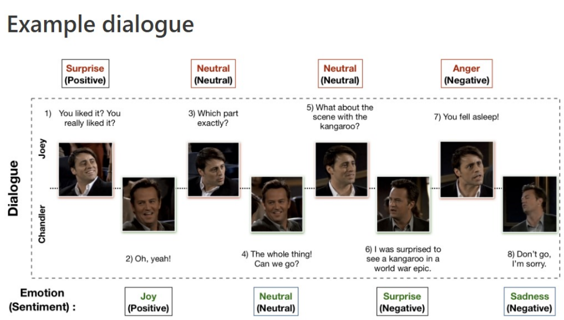

Fig. 1. Example dialogue from MELD dataset showing utterance-level annotations.

B. Pre-processing

1) Text Pre-processing: Text utterances were preprocessed using the following pipeline:

- Lowercasing all characters for uniformity
- Tokenization using the bert-base-uncased tokenizer, which internally handles punctuation and casing.
- Tokenizing using the `bert-base-uncased` tokenizer
- Truncating or padding token sequences to a maximum length of 64 tokens
- Generating attention masks to distinguish padded and valid tokens

These steps ensure compatibility with the BERT encoder while preserving sentence semantics.

2) Audio Pre-processing: Audio pre-processing involved aligning audio clips with their corresponding utterances using metadata from the MELD `.yaml` files:

- Audio clips were extracted using `FFmpeg` from original video recordings
- Sampling rate was standardized to 16 kHz
- Silence and background noise were minimally filtered to preserve emotion cues
- Log-mel spectrograms were generated using Whisper's log-mel spectrogram front-end; Whisper does not use a tokenizer for audio but instead converts the waveform directly into a sequence of acoustic features

Given the computational intensity of Whisper preprocessing, training time and memory usage are higher than traditional audio pipelines.

C. Spectrogram-Based Audio Model: Vision Transformer (ViT)

To explore emotion recognition from acoustic signals using interpretable representations, we first implemented a model based on mel-spectrogram inputs. Raw audio files were preprocessed by trimming silence and generating mel-spectrograms using the `Librosa` library. Each spectrogram was normalized and resized to a fixed dimension of 256×256 to ensure consistent input shape across utterances of varying durations. These 2D spectrograms were treated as grayscale images and fed into a Vision Transformer (ViT) architecture tailored for single-channel input.

The ViT model adopted a patch-based approach, splitting each spectrogram into non-overlapping patches of size 8×8, yielding a sequence of flattened vectors. Each patch embedding was passed through a positional encoder before being input into a series of transformer layers. The model configuration included six transformer blocks with four attention heads per layer, a hidden dimension of 512, and a feedforward projection size of 2048. A class token was prepended to the patch sequence and used for final classification into one of the seven emotion categories via a linear head.

This model was trained using the Adam optimizer and a learning rate of 1×10^{-4} with a step decay scheduler. Cross-entropy loss was used as the objective function, and early stopping was considered based on validation performance. The spectrogram-based ViT served as a mid-level complexity baseline, enabling us to evaluate

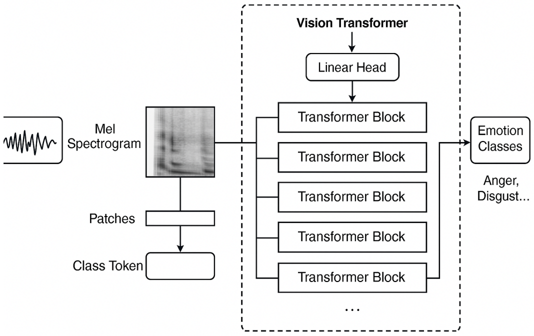

Fig. 2. ViT model for mel-spectrogram input

the discriminative power of frequency-domain features before transitioning to waveform-level encoders.

D. Text Encoder: BERT

After preprocessing, the dataset was divided into training, validation, and test splits. The training set was used to fine-tune a BERT-based model for the task of emotion recognition.

For the textual modality, we employed the `bert-base-uncased` model from HuggingFace as the backbone of our classification pipeline. This architecture consists of 12 transformer encoder layers, each with 12 self-attention heads and a hidden size of 768. The model was implemented using the `BertForSequenceClassification` class, configured for seven emotion categories. A classification head was appended to the BERT encoder and trained in an end-to-end fashion.

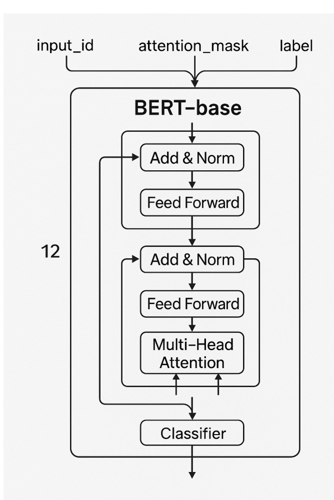

Fig. 3. BERT Model Architecture

The training procedure leveraged the `outputs['loss']` value from BERT's forward pass as the objective function. Optimization was carried out using the AdamW optimizer with grouped weight decay, and a learning rate of 1×10^{-5} was applied with a linear warmup schedule. To ensure robust generalization and prevent overfitting, early stopping was based on the validation F1-score, with model checkpoints saved whenever performance improved.

During training, the model received token IDs, attention masks, and corresponding emotion labels. Standard backpropagation was followed by gradient clipping to enhance stability. At the conclusion of training, the best model based on validation performance was restored for final evaluation. This training strategy enabled the model to capture deep contextual dependencies within utterances, effectively leveraging BERT's representational power for accurate emotion classification.

E. Audio Encoder: Whisper

The audio encoder was built upon the Whisper architecture developed by OpenAI. Instead of using Whisper for automatic speech recognition, we adapted it to function as a feature extractor. The encoder was initialized with pretrained Whisper weights and repurposed to extract acoustic representations from individual utterances. To reduce computational complexity and ensure stability, the encoder operated in a frozen state throughout training, without fine-tuning.

Audio inputs were first resampled and transformed into log-mel spectrograms using Whisper's built-in preprocessing utilities. These spectrograms were then processed through the Whisper encoder, which consists of two convolutional layers followed by six stacked residual attention blocks. The encoder produced a rich sequence of acoustic feature vectors, which served as the input to a custom neural architecture designed for classification.

Following the Whisper encoder, the feature sequence was passed through an additional 1D convolutional block comprising a `Conv1d(1500` $\rightarrow$ `128)` layer, batch normalization, ELU activation, max pooling, and dropout. This stage helped reduce dimensionality while introducing non-linear transformations and regularization.

The output from the convolutional block was flattened across spatial dimensions to produce a high-dimensional vector. This vector was then passed into a compressor block implemented as a fully connected linear layer of size `Linear(16384` $\rightarrow$ `1024)`, followed by ELU activation, batch normalization, and dropout. This step significantly reduced the representation size while preserving informative features.

Subsequently, the compressed embedding was passed through a multi-layer perceptron (MLP) classifier consisting of three fully connected layers. The first layer mapped the input from 1024 to 256 dimensions, followed by ELU activation, dropout, and batch normalization. The second layer maintained the dimensionality at 256 and applied the same operations. The final layer reduced the embedding to a 64-dimensional vector, again followed by ELU and dropout.

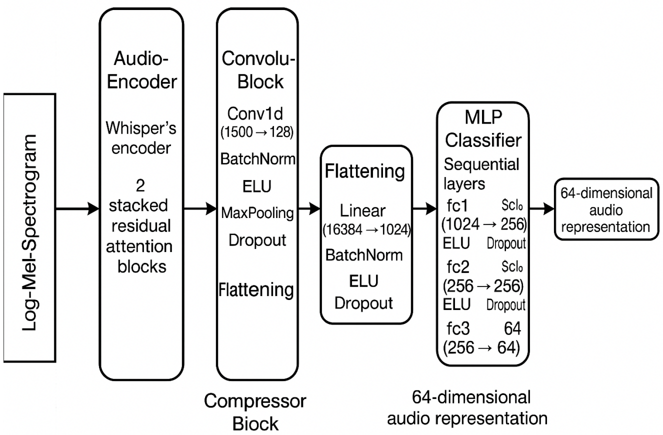

Fig. 4. Whisper Model Architecture

This final 64-dimensional audio embedding was used as input to the multimodal fusion model, where it was concatenated with the textual representation. The overall architecture thus retained Whisper's robust acoustic encoding capabilities while enabling efficient classification in a modular and scalable manner.

F. Fusion Model

Our multimodal architecture adopts a **late-fusion strategy** to integrate audio and text modalities for emotion classification. In this approach, we first extract a 768-dimensional text embedding from BERT by using the [CLS] token and a 64-dimensional audio embedding from the custom Whisper-based encoder. These two modality-specific vectors are then concatenated to form an 832-dimensional joint representation.

This fused vector is passed through a multi-layer feedforward classifier designed to progressively reduce dimensionality while applying non-linearity and regularization. The architecture of the classifier comprises four key layers:

- **fc1**: Linear(832 → 1024) followed by ELU activation, Dropout(0.2), and Batch Normalization
- **fc2**: Linear(1024 → 256), ELU, Dropout(0.2), BatchNorm
- **fc3**: Linear(256 → 64), ELU, Dropout(0.2)
- **classifier**: Linear(64 → 7), which maps the final embedding to one of the seven emotion categories

To ensure robustness and fairness in classification, especially for underrepresented emotion classes, we employed a class-weighted `CrossEntropyLoss`. This helped mitigate class imbalance and improve the model's sensitivity to minority classes. Additionally, model performance was evaluated using the **weighted F1-score**, providing a balanced measure across all emotion categories.

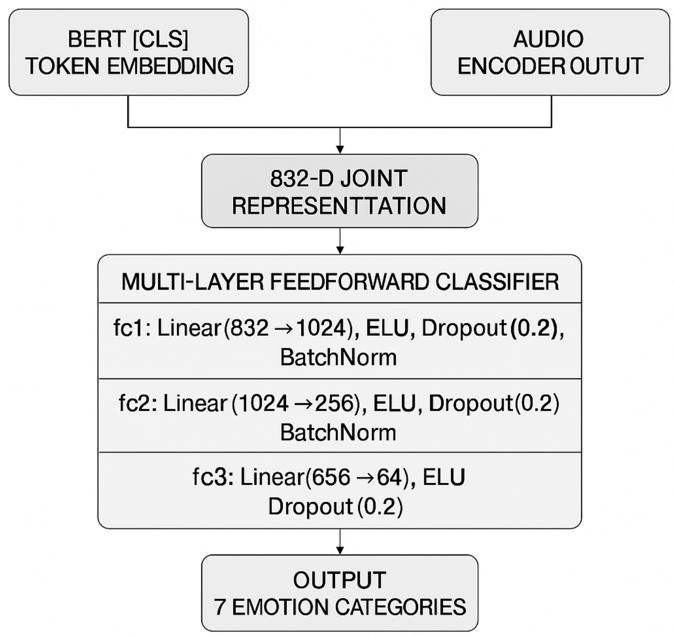

Fig. 5. Fusion Model Architecture

Overall, this late-fusion design maintains the unique characteristics of each modality while enabling the classifier to learn joint representations for more effective and generalized emotion recognition.

IV. Model Building

The model development process followed a structured progression, beginning with a simple unimodal baseline and culminating in a multimodal architecture that jointly leverages acoustic and textual cues. Each model was selected based on its capacity to overcome specific limitations identified in preceding approaches. This section details the rationale behind each modeling choice, highlighting the evolution of architectural complexity and the trade-offs considered at each stage.

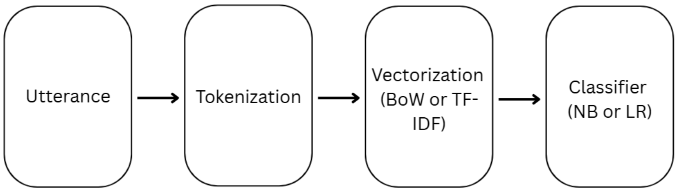

Fig. 6. NLP pipeline for emotion classification

A. Baseline Model: TF-IDF with Naive Bayes and Logistic Regression

To establish an initial benchmark, we implemented classical NLP models using TF-IDF and bag-of-words representations in combination with Naive Bayes and Logistic Regression classifiers. These models served as a fast and interpretable baseline, offering meaningful insights before progressing to more complex architectures like BERT and Whisper.

The rationale for starting with these models was threefold. First, their interpretability allowed us to identify which words or n-grams contributed most to predictions, offering transparency into early feature behavior. Second, they provided a useful performance floor, helping to quantify the relative gains achieved by modern deep learning methods. Finally, their low computational cost made them ideal for rapid prototyping and early validation of dataset quality and label consistency.

In terms of feature extraction, two vectorization strategies were explored. For Naive Bayes, we used a `CountVectorizer` with tokenization based on NLTK's `word_tokenize` function. For Logistic Regression, we employed a `TfidfVectorizer` with an n-gram range of (1,5) and a cap of 1000 features. This configuration was selected to balance local and global context while preventing overfitting and limiting sparsity. Importantly, we avoided stop-word removal to preserve negations and affect-laden terms critical to emotion classification, such as "not", "never", or "hate".

Naive Bayes was chosen for its alignment with count-based features and its probabilistic treatment of word occurrences under a conditional independence assumption. In contrast, Logistic Regression models the log-odds of class membership and complements the TF-IDF representation well, especially when learning from weighted, real-valued features.

To streamline experimentation, we used `sklearn.pipeline.Pipeline` to encapsulate each model's preprocessing and classification steps. This design ensured consistent transformations across training and test data and minimized the risk of data leakage. Additionally, it allowed quick model switching and clean integration of evaluation steps.

This baseline setup was critical in establishing early confidence in the dataset and defining performance expectations for the more advanced architectures discussed in subsequent sections.

B. Spectrogram-Based Audio Model

To explore the standalone effectiveness of acoustic features, we constructed a model that operates purely on mel-spectrogram representations derived from audio recordings. The core motivation behind this approach was to examine how far nonverbal cues such as pitch contours, harmonic energy, and spectral shifts could be leveraged in isolation for emotion classification. Unlike raw waveform-based encoders or ASR-centric feature extractors like Whisper, this method emphasizes interpretable, handcrafted representations as input to a data-driven deep learning model.

We selected mel-spectrograms due to their perceptual alignment with human hearing and their widespread success in emotion recognition literature. These spectrograms were generated by applying short-time Fourier transforms followed by mel-scale filter banks and log compression. To ensure uniformity and compatibility with transformer-based input schemes, each spectrogram was padded or truncated to a fixed shape of 256×256, with amplitude values normalized to the $[-1, 1]$ range. This transformation ensured consistent dimensionality across variable-length utterances, allowing the model to focus on frequency and temporal structure rather than input length.

The architecture for this model was centered around a Vision Transformer (ViT) adapted for single-channel input. We divided each spectrogram into non-overlapping patches of size 8×8, resulting in 1024 patches per image. A learnable class token was prepended to this patch sequence, enabling the transformer to aggregate global context and make holistic predictions. Each patch was flattened and linearly projected to a fixed embedding dimension of 512, with positional embeddings added to retain spatial coherence.

The transformer backbone consisted of 6 self-attention blocks, each with 4 heads and feedforward dimensions of 2048, offering a balance between capacity and overfitting control. Unlike CNNs, ViT does not encode strong spatial locality priors, which allowed the model to flexibly learn which spectral regions—temporal or frequency-based—were relevant for emotion detection. This property was particularly beneficial in dealing with highly variable prosodic patterns across emotions.

Regularization played a critical role in this setup. We applied dropout both in the embedding and attention layers, and incorporated layer normalization after each sublayer to stabilize training. The model was optimized using Adam with a small learning rate and a step-based scheduler to encourage convergence while avoiding oscillations. Cross-entropy loss was used with balanced class weights to counteract label imbalance in the MELD dataset.

Beyond classification performance, one of the advantages of this approach was interpretability. Attention maps from the transformer layers could be visualized to identify which frequency bands and time intervals the model relied on most for emotion prediction. This interpretability offers potential for future applications in clinical or educational settings where transparency is critical.

Ultimately, the spectrogram-based model served as a valuable midpoint between traditional signal-processing pipelines and large pretrained models. It demonstrated that with sufficient architectural design and normalization, spectrograms alone can encode emotion-relevant acoustic information without the need for extensive pretraining or multimodal context.

C. Bert Fine Tuning

To move beyond shallow lexical representations, we adopted a transformer-based architecture capable of modeling context-dependent semantics for emotion classification. The `bert-base-uncased` model was chosen not only for its state-of-the-art performance on a variety of NLP benchmarks but also for its proven ability to capture nuanced emotional cues in sentence-level discourse—an essential requirement given the conversational nature of the MELD dataset.

The fine-tuning pipeline was designed with an emphasis on generalization, computational efficiency, and modularity. Utterances were tokenized using BERT's subword tokenizer, with each sequence truncated or padded to a fixed maximum length of 128 tokens. This length was selected to preserve meaningful content while maintaining GPU memory efficiency.

To ensure effective regularization, model parameters were grouped to apply weight decay selectively: bias terms and normalization layers were excluded from decay, consistent with best practices that suggest unconstrained optimization benefits these parameters. Training optimization was performed using the AdamW optimizer with a learning rate of 1×10^{-5}, and a linear learning rate scheduler with warm-up steps was employed to stabilize gradient updates during early epochs.

Emotion labels were encoded using ordinal encoding to support efficient computation with PyTorch's loss functions. Attention masks were automatically generated to direct the model's focus to non-padding tokens during both forward and backward passes, eliminating the need for manual sequence alignment.

Several additional safeguards were incorporated to mitigate overfitting. Early stopping was applied based on the macro-averaged F1-score, which better captures balanced performance across the seven emotion classes particularly important in the presence of class imbalance. Gradient clipping was also employed to prevent unstable parameter updates during training. Model checkpoints were saved at the highest validation F1-score to ensure robust downstream evaluation.

Overall, these design choices enabled BERT to serve not merely as a pretrained feature extractor, but as a finely tuned encoder capable of leveraging deep contextual information for emotion classification in multimodal, dialogue-rich settings.

D. Whisper-Based Audio Model

In moving from lexical to acoustic modeling, our goal was to capture non-verbal emotional cues such as pitch, intonation, and energy that often complement or disambiguate textual sentiment. Whisper, originally designed for multilingual automatic speech recognition, offers strong general-purpose audio representations thanks to its convolutional and transformer-based encoder stack. Rather than using it as a black-box frozen extractor, we opted to fine-tune its weights on emotion classification. This decision was driven by the intuition that emotion-rich vocal patterns differ significantly from general ASR tasks, and adaptation was needed for optimal performance.

We leveraged Whisper's internal log-mel spectrogram pipeline to ensure compatibility with its pretrained parameters. Instead of treating the encoder's output as a sequence, we summarized temporal dynamics via the mean and standard deviation across time — a design choice balancing expressiveness and memory efficiency. This representation preserves speaker-dependent dynamics while reducing variability across utterance lengths, allowing downstream classifiers to operate on fixed-length vectors.

To further shape these features into emotion-discriminative embeddings, we applied a convolutional transformation followed by a compression layer and a deep MLP. This layered setup was motivated by two key observations: (1) raw Whisper features, though informative, are high-dimensional and sparse for small datasets; and (2) deep projections help uncover latent structure while allowing flexible non-linear decision boundaries.

We also introduced normalization, dropout, and ELU activation at each stage to address overfitting — a risk when fine-tuning large encoders on medium-sized datasets. Class imbalance was mitigated using weighted loss functions, and audio features were standardized using global batch statistics prior to training. Unlike handcrafted prosodic features or shallow MFCC-based models, this approach offered a unified, data-driven pipeline for acoustic emotion recognition that could adaptively reweight low-level and high-level cues as needed.

E. Multimodal Fusion Model

To improve emotion recognition beyond single-modality models, we combined the strengths of both text and audio features using a late fusion approach. This decision was based on our observation that while BERT and Whisper performed well individually, they often failed in different situations. The text model struggled with short or ambiguous phrases, while the audio model had difficulty with flat or emotionless speech. By combining the two, we aimed to create a more balanced and accurate system.

In this fusion model, we extracted a 64-dimensional feature vector from Whisper and a 768-dimensional feature from BERT's [CLS] token, then concatenated them to form a single 832-dimensional vector. This combined feature was passed through a series of fully connected layers with batch normalization, ELU activation, and dropout. These layers helped the model learn useful patterns from both text and audio, while also reducing overfitting.

We chose concatenation instead of averaging because it allows the model to keep the unique information from each modality. We also used a weighted cross-entropy loss function to handle class imbalance, especially for emotions like *fear* or *disgust*, which were less common in the dataset. The model was trained using the Adam optimizer with a learning rate scheduler and gradient clipping for stability.

Unlike early fusion methods, which try to align audio and text at the token or frame level, our late fusion model avoids synchronization issues. It treats each modality independently at first and combines them only after they have been processed, making the design more flexible and easier to train.

Overall, the fusion model worked as a joint decision-maker. Rather than simply averaging two predictions, it learned how to use information from both BERT and Whisper together. This helped it achieve better performance than either model alone, especially in cases where one modality lacked clear emotional signals.

V. Results

To evaluate model performance on the MELD dataset, we focus on two primary metrics: accuracy and weighted avg score. While accuracy provides an overall measure of correct classifications, the weighted F1-score offers a more robust evaluation in the presence of class imbalance by considering both precision and recall across emotion categories in proportion to their frequency. This is especially important in MELD, where the neutral class dominates, and minority classes such as fear and disgust are underrepresented.

In addition to these quantitative metrics, we analyze confusion matrices to uncover patterns of misclassification. These visualizations highlight specific challenges in distinguishing between acoustically or semantically similar emotions and offer insight into each model's sensitivity to different emotional expressions. Together, these evaluation tools enable a nuanced understanding of the relative strengths and limitations of each model architecture from traditional baselines to deep learning and multimodal fusion approaches.

A. Emotion Classification Using Spectrogram-Based ViT

To evaluate the standalone capability of acoustic features for emotion recognition, we trained a Vision Transformer (ViT) model using mel-spectrogram representations of the MELD dataset's audio utterances. These spectrograms encode frequency content over time in a 2D format, allowing the Vision Transformer to identify visual patterns in the time-frequency domain that may indirectly correspond to speech characteristics associated with emotional expression.

Figure 7 presents the mini-batch loss fluctuations recorded during early iterations of training. The x-axis represents the number of iterations, while the y-axis reflects the corresponding mini-batch loss values. Initially, we observe a rapid drop in loss, suggesting that the model begins to learn useful patterns from the acoustic input. However, after the first few updates, the loss values plateau and begin to fluctuate narrowly around a relatively high level (approximately 2.4). This plateau indicates that the model has reached a point where it is no longer significantly improving, despite continued training.

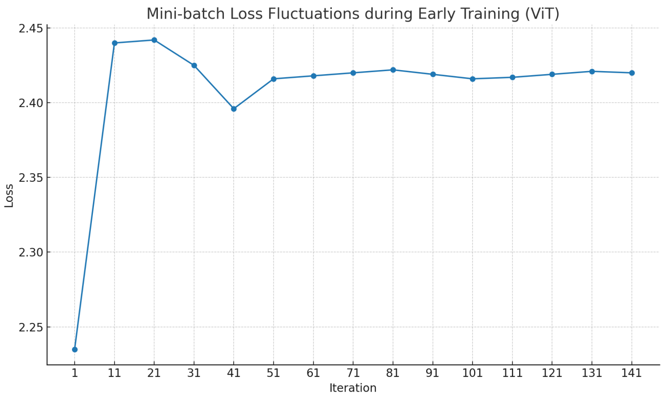

Fig. 7. Mini-Batch Loss Fluctuations During Early Training (ViT)

Importantly, while the loss showed early convergence, the model's classification performance as measured by accuracy on both training and validation sets remained very low throughout the process. This poor performance revealed a critical limitation: the mel-spectrogram inputs did not provide enough discriminatory information across emotion classes. Upon closer inspection of the dataset, we found that many utterances shared similar image variations across emotions such as "neutral", "sadness", and "anger", making it difficult for the model to reliably distinguish between them based on nonverbal cues alone.

This lack of variation in the dataset severely constrained the model's learning capacity. While the ViT architecture is theoretically well-suited to extracting spatial patterns, its effectiveness is ultimately tied to the richness and diversity of the input data. In this case, the uniformity of audio images across different emotional states resulted in high confusion between classes and poor generalization to validation data.

As a result of these findings, we concluded that a unimodal approach based solely on spectrogram images was insufficient for the task of emotion recognition. The limitations observed in the ViT model's performance motivated us to transition toward a multimodal modeling strategy, where we incorporate both spectrogram images and textual features of the utterances. By doing so, we aim to leverage the complementary strengths of multi-faceted information to improve classification accuracy and overall model robustness.

1) Preprocessing Considerations and Technical Challenges: The preprocessing pipeline involved silence trimming, amplitude normalization, and resizing of mel-spectrograms to a fixed resolution of 256×256. This was essential for ensuring consistent input dimensions. However, short utterances occasionally triggered `n_fft` warnings during spectrogram generation due to insufficient signal length. These cases were handled through zero-padding to maintain consistent time-frequency resolution.

The dynamic range of spectrogram values was normalized to the $[-1, 1]$ range before feeding into the model. This standardization helped the model focus on relative frequency activations rather than raw amplitude, which varies significantly across speakers and recordings.

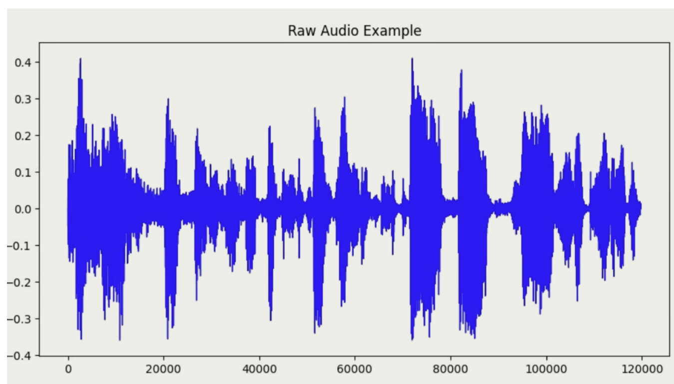

Fig. 8. Waveform of a MELD utterance used as input to the spectrogram-based ViT model

2) Visual Interpretation of Input Representations: To assess the quality and consistency of acoustic inputs, we visualized several representative spectrograms alongside their corresponding raw audio waveforms. While these plots confirmed that basic prosodic features such as pitch and intensity were captured, they also revealed a notable lack of distinctive variation across many emotion classes. In particular, the spectral energy distributions of emotions like "neutral," "sadness," and "anger" appeared visually similar, especially in shorter utterances. This overlap likely contributed to the model's confusion between classes. Figure 9 presents examples of spectrogram inputs, illustrating how subtle or ambiguous differences in time-frequency profiles limited the model's ability to make reliable distinctions based on acoustic features alone.

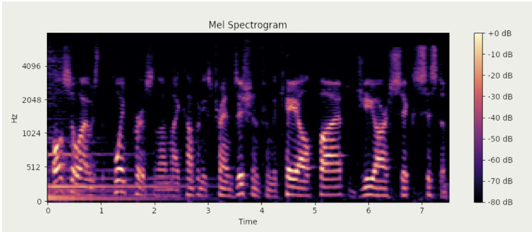

Fig. 9. Mel-spectrogram of a spoken utterance from the MELD dataset

The spectrogram-based ViT model revealed that relying solely on acoustic features for emotion recognition yields limited performance, primarily due to the low variability in the dataset's audio signals. The model failed to achieve satisfactory accuracy on both training and validation sets. This consistent underperformance highlights that acoustic-only representations—particularly those derived from short, emotionally subtle utterances lack the richness needed to reliably distinguish between overlapping or nuanced emotion categories. These findings suggest that acoustic features alone are insufficient for robust or fine-grained emotion classification, especially in real-world applications where precision and contextual understanding are critical.

B. Text-Only Baselines for Emotion Recognition

To establish benchmark performance using text alone, we evaluated two classical models: a Naive Bayes classifier with a bag-of-words representation and a Logistic Regression model using TF-IDF features. Both models were trained and tested on MELD utterances without any acoustic or contextual features.

1) Classification Performance of TF-IDF + Logistic Regression: To establish a more competitive text-only baseline, we trained a Logistic Regression classifier on TF-IDF features extracted from MELD utterances. The model utilized both unigrams and bigrams and incorporated class weighting (`class_weight="balanced"`) to mitigate the effects of class imbalance. Evaluation was conducted on a held-out test set containing 2,610 utterances.

	precision	recall	f1-score	support
negative	0.56	0.49	0.52	833
neutral	0.71	0.71	0.71	1256
positive	0.48	0.58	0.53	521
accuracy			0.61	2610
macro avg	0.59	0.59	0.59	2610
weighted avg	0.62	0.61	0.62	2610

Fig. 10. Classification report for Logistic Regression using TF-IDF features

The model achieved an overall accuracy of 61%, a macro-averaged F1-score of 0.59, and a weighted avg score of 0.62. Class-wise performance showed a clear advantage for the *neutral* class, with precision, recall, and F1-score all reaching 0.71 a reflection of the lexical clarity and class dominance of neutral utterances in the dataset.

In contrast, the *positive* class achieved an F1-score of 0.53, with precision at 0.48 and recall at 0.58, suggesting overprediction in emotionally ambiguous cases. The *negative* class recorded a precision of 0.56, but lower recall at 0.49, yielding an F1-score of 0.52, indicating a tendency to under-recognize negative sentiments. These findings suggest the model captures dominant sentiment cues well but struggles with subtleties or overlap between sentiment categories.

2) Confusion Matrix Analysis: The raw and normalized confusion matrices offer critical insights into the Logistic Regression model's class-wise behavior, revealing systematic patterns of misclassification that align with the challenges of text-only emotion recognition.

As illustrated in Figure 11, the neutral class exhibited the strongest diagonal concentration, with 892 out of 1,256 instances correctly predicted. This reflects the model's proficiency in recognizing lexically consistent, emotionally neutral language often characterized by filler words, affirmations, or noncommittal responses. In contrast, predictions for the positive and negative classes revealed substantial off-diagonal activity. For example, of the 833 negative instances, only 410 were correctly classified, with the rest split between neutral and positive predictions. Similarly, the positive class saw 303 correct predictions out of 521, with misclassifications largely toward the neutral category.

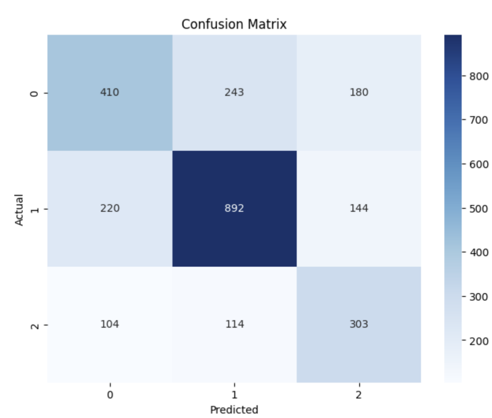

Fig. 11. Confusion Matrix

The normalized confusion matrix (Figure 10) makes these tendencies even more evident. Approximately 29% of negative utterances were misclassified as neutral, and 22% of neutral utterances were labeled as positive. Additionally, about 20% of positive samples were mistaken as negative, indicating that the model struggles to clearly distinguish between opposing sentiments in the absence of strong polarity cues.

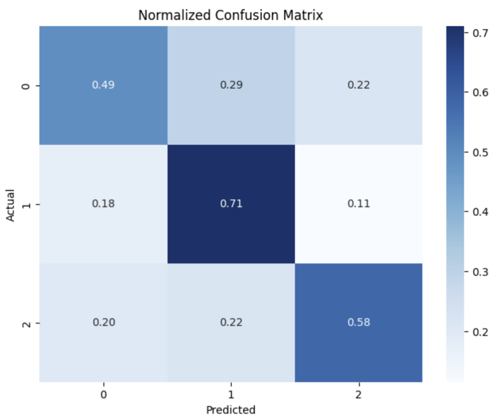

Fig. 12. Normalized Confusion Matrix

These misclassifications can largely be attributed to three factors. First, the lexical overlap between sentiment categories many emotionally nuanced utterances use similar vocabulary regardless of class. Second, the lack of intonation, emphasis, and speaker intent deprives the model of prosodic signals that would otherwise help differentiate sarcasm, irony, or subtle disapproval. Third, utterance brevity in MELD often means that emotional context is implicit or distributed across a dialog, making it inaccessible to a classifier trained on single-sentence inputs.

Despite these limitations, the model consistently predicted the majority class (neutral) with high reliability, and its misclassification patterns were not random but concentrated around adjacent sentiment categories. This behavior, while imperfect, reflects a logical decision boundary given the absence of multimodal or contextual features. Consequently, the Logistic Regression model still serves as a strong interpretability-oriented benchmark for text-only settings, especially in low-resource environments where speed and transparency may outweigh marginal accuracy gains.

C. *Emotion Classification with BERT (Text-Only)*

To evaluate the performance of a deep contextual language model on emotion classification, we fine-tuned a pretrained BERT-base model on MELD utterances labeled with seven emotion categories. Unlike traditional models, BERT is capable of modeling complex syntactic and semantic dependencies, making it particularly suited for capturing subtle emotional cues present in conversational text.

Following fine-tuning, the model achieved a macro F1-score of 0.41, with notable performance on well-represented classes such as joy, anger, and sadness. However, it continued to struggle with underrepresented or less linguistically explicit classes like fear and surprise. These results highlight both the strength of transformer-based architectures in learning from emotionally charged language, and the ongoing challenges posed by class imbalance and context-dependent expressions in spoken dialog.

1) Training Dynamics and Optimization: Training was conducted over a maximum of 10 epochs using the AdamW optimizer with a learning rate of 1×10^{-5}, coupled with a linear learning rate scheduler to adjust the learning rate dynamically over time. A batch size of 32 and a maximum sequence length of 128 tokens were used. To prevent overfitting, an early stopping mechanism with a patience of 2 epochs was employed, monitoring the macro F1-score on the validation set after each epoch.

Throughout training, the model demonstrated stable convergence, with the training loss steadily decreasing from 1.39 in epoch 1 to 0.76 by epoch 7. Validation performance, as measured by macro-averaged F1-score, rose sharply in the early epochs from 0.30 at the start to 0.41 by epoch 5, which marked the best performing point during training. Following this, two successive epochs failed to improve the F1-score further, prompting early stopping at epoch 7. The best model checkpoint from epoch 5 was subsequently restored for final evaluation.

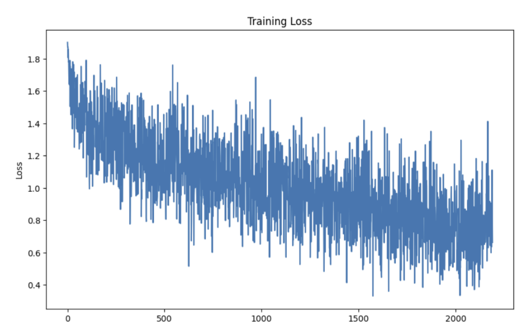

Fig. 13. Training loss curve over 10 epochs

The full training loss progression is illustrated in

Figure 13, showing a smooth decline with no signs of instability or overfitting. This learning behavior highlights the effectiveness of using pretrained contextual embeddings for emotion classification in text.

2) Performance Metrics: Final evaluation was conducted on the MELD validation set consisting of 1,109 utterances. The BERT model achieved an overall accuracy of 60%, a macro-averaged F1-score of 0.41, and a weighted F1-score of 0.57. The weighted F1-score, which accounts for class imbalance by assigning more weight to prevalent classes, provides a more representative summary of the model's overall performance than the macro average alone.

These results demonstrate moderate performance, with significantly better recall and precision for majority classes and noticeable degradation on low-frequency emotion categories. Class-wise metrics are summarized in Table 1. The model performed best on the *joy* class, achieving an F1-score of 0.76, supported by high precision (0.71) and recall (0.82). This strong performance can be attributed to both the expressive nature of joyful language and the high number of training samples available for this class. Similarly, the *anger* class saw relatively strong results, with an F1-score of 0.59.

The *sadness* class also achieved a balanced F1-score of 0.53, with a precision of 0.49 and recall of 0.59, indicating the model was reasonably effective at detecting sad emotional cues. In contrast, the model struggled to generalize on minority classes such as *fear* and *surprise*, which had limited training support and more subtle or context-dependent linguistic markers. The *fear* class had a recall of just 3%, while *surprise* reached only 14%, indicating that many samples were misclassified as *neutral* or other dominant emotions.

TABLE I
CLASS-WISE PERFORMANCE METRICS OF THE BERT MODEL ON THE MELD VALIDATION SET.

Emotion Label	**Precision**	**Recall**	**F1-score**	**Support**
Neutral (0)	0.43	0.37	0.40	153
Surprise (1)	0.60	0.14	0.22	22
Fear (2)	0.50	0.03	0.05	40
Sadness (3)	0.49	0.59	0.53	163
Joy (4)	0.71	0.82	0.76	470
Disgust (5)	0.44	0.24	0.31	111
Anger (6)	0.57	0.62	0.59	150
Macro avg	0.53	0.40	0.41	1109
Weighted avg	0.58	0.60	0.57	1109
Overall (avg)	—	—	—	1109

This disparity in performance across classes reinforces the impact of class imbalance, as well as the limitations of relying on text-only input for fine-grained emotion detection. Emotions that are less frequently expressed or contextually nuanced are more likely to be confused with dominant or affectively flat categories like *neutral*.

3) Confusion Matrix Insights: The confusion matrix shown in Figure 14 highlights both the strengths and limitations of the BERT-based emotion classifier. Clear diagonal dominance is observed for well-represented classes such as joy, neutral, and anger, indicating that the model was consistently able to predict these emotions correctly when they appeared in the test data. In particular, the joy class showed high recall and precision, reaffirming the model's ability to recognize strongly positive affective signals, especially when they are explicitly stated in text.

However, the matrix also reveals substantial misclassification across emotionally adjacent or lexically similar categories. One of the most notable patterns involves the fear class, which was frequently misclassified as either neutral or sadness. This suggests that the model struggles to distinguish fear-related expressions when they are subtle, implicit, or context-dependent common traits in human emotional communication. Similarly, surprise often leaked into other categories due to its ambiguous tone and sparse representation in the training data.

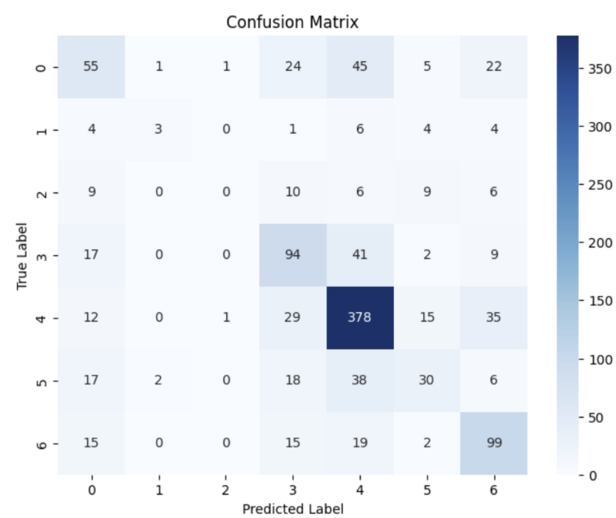

Fig. 14. Confusion Matrix for Bert Model

These confusions underscore a fundamental limitation of text-only emotion recognition: the absence of acoustic, prosodic, or dialog context restricts the model's capacity to interpret non-literal language, emotional nuance, and pragmatic shifts. BERT's contextual embeddings, while powerful, are still bounded by the semantic signal available in a single utterance. When emotional expression is implied through tone, hesitation, irony, or interaction history, a text-only model lacks the multimodal grounding to resolve such ambiguity.

Thus, although the model demonstrates strong predictive accuracy on dominant and emotionally explicit classes, the confusion matrix clearly illustrates the need for complementary modalities particularly audio or dialog history for robust and comprehensive emotion understanding.

D. Emotion Classification Using Whisper Audio Embeddings

To evaluate the standalone contribution of acoustic cues in emotion classification, we developed a speech-based model pipeline leveraging high-dimensional embeddings extracted from a fine-tuned Whisper encoder. The Whisper model, originally trained for robust automatic speech recognition (ASR), encodes rich prosodic, rhythmic, and spectral information across spoken utterances. Rather than relying on transcribed text, we used its intermediate encoder outputs as dense representations of raw audio, capturing features such as pitch variation, energy contours, pause patterns, and spectral dynamics.

These audio embeddings, generated for each MELD utterance, served as the sole input to a series of downstream classifiers. Importantly, the model architecture excluded any access to textual data or dialog history, isolating the predictive capacity of nonverbal acoustic information. This approach allowed us to systematically assess how much emotional content can be inferred purely from voice tone, intensity, and prosody, without relying on lexical semantics or contextual dialogue cues.

By focusing exclusively on acoustic signals, this experiment also provides a critical baseline for subsequent multimodal fusion analysis. It tests whether Whisper-derived representations can support emotion recognition independently, especially in cases where text is unavailable, ambiguous, or insufficient to capture affective nuance.

1) Training Overview and Optimization: To explore how acoustic representations from Whisper embeddings could be used to classify emotions, we designed and evaluated two downstream classifier architectures. The first was a feedforward neural network (MLP) that operated on a 1024-dimensional pooled feature vector, obtained by averaging the temporal outputs of the Whisper encoder across time. This global summarization approach prioritized overall spectral trends rather than localized fluctuations. The second architecture preserved the temporal structure of Whisper outputs shaped as $[B, 1500, 512]$, which were passed through a stack of 1D convolutional layers followed by batch normalization, max pooling, and dense projection. This allowed the model to extract localized acoustic features such as rising pitch, energy bursts, and rhythmic shifts signals often associated with affective speech patterns.

Both models were trained using class-weighted cross-entropy loss to counteract label imbalance, with a batch size of 8 and Adam optimizer for parameter updates. A linear learning rate scheduler with warmup was employed to stabilize early training. The training process spanned 10 epochs, and a validation-based checkpointing strategy was implemented to preserve the model yielding the highest macro F1-score on the validation set.

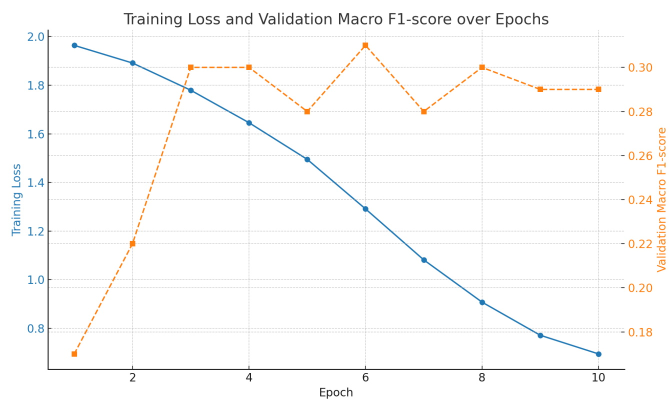

Fig. 15. Training loss and validation macro F1-score across epochs

Throughout training, the models exhibited smooth and consistent convergence. The training loss steadily declined from 1.96 in epoch 1 to 0.69 by epoch 10, with no signs of plateauing or instability. This suggests that Whisper embeddings captured sufficient acoustic information for the classifiers to generalize across emotion classes. The learning curve clearly reflects effective optimization, reinforcing the viability of acoustic-only inputs for emotion classification even in the absence of text or dialog context.

2) Model Selection via Checkpoint Averaging: To improve generalization and stabilize model predictions, we employed a checkpoint ensembling strategy. After training for 10 epochs, we selected the three best-performing checkpoints based on their validation weighted average F1-score: `epoch_2` (w_avg_score = 0.40), `epoch_5` (w_avg_score = 0.47), and `epoch_3` (w_avg_score = 0.43). We then computed the average of their weights using a custom PyTorch utility that performs elementwise summation followed by normalization across corresponding parameters. The resulting averaged state dictionary was used to initialize the final model for downstream evaluation.

This achieved an **overall accuracy of 36%**, a **Weighted averaged score of 0.48**, and a **weighted F1-score of 0.35**. These results represent a slight improvement over single-checkpoint performance, suggesting that the averaging strategy improved robustness across classes.

Class-wise evaluation reveals that the model performed reasonably well on high-sample or acoustically distinct classes. For instance, *anger* achieved an F1-score of 0.45, supported by a relatively high recall of 0.57, indicating that the model captured its prosodic intensity effectively. Similarly, *neutral* was predicted with a strong precision of 0.66, although its recall was lower at 0.27 suggesting that the model frequently defaulted to this class in ambiguous or unclear cases.

In contrast, performance on minority and affectively subtle classes was considerably lower. The *disgust* class yielded an F1-score of only 0.07, primarily due to its extremely low precision (0.04) and sparse representation in the training set. The *fear* class also exhibited weak predictive performance, with an F1-score of 0.15 despite a recall of 0.25. These results indicate that Whisper embeddings alone may not encode sufficient discriminative information for reliably distinguishing subtle or underrepresented emotions.

TABLE II
PERFORMANCE OF THE AVERAGED WHISPER-BASED MODEL ($n = 1{,}108$).

Emotion	Precision	Recall	F1-score	Support
Anger	0.41	0.65	0.50	153
Disgust	0.05	0.05	0.05	22
Fear	0.17	0.12	0.14	40
Joy	0.27	0.47	0.34	163
Neutral	0.72	0.21	0.32	469
Sadness	0.32	0.41	0.36	111
Surprise	0.28	0.49	0.36	150
Overall	—	—	—	1,108
Macro Avg	0.32	0.34	0.30	—
Weighted Avg	0.48	0.36	0.35	—

Interestingly, *surprise* achieved a relatively high F1-score of 0.37 comparable to that of *anger* and *sadness* highlighting the model's ability to recognize emotionally expressive and intonation-rich speech patterns. This supports the value of prosodic modeling for certain types of emotions. However, the overall variability in class-wise performance underscores the limitations of audio-only input for comprehensive emotion classification in natural, conversational speech.

3) Test Set Results: On the final held-out **test set** (n = 2,609), the Whisper-based classifier achieved a **Weighted avg Score of 0.47**, and a **weighted F1-score of 0.36**. These results are closely aligned with those observed on the validation set, indicating that the model's performance generalized consistently to unseen data. Although absolute performance remains modest, this consistency across datasets underscores the model's capacity to extract stable, prosody-based features from audio inputs.

Class-wise results revealed that the model was most effective at identifying emotions that are often accompanied by expressive intonation or distinct vocal patterns. The *anger* class achieved a score of 0.32, supported by relatively strong recall, suggesting the model's sensitivity to elevated pitch, speech rate, or volume typical prosodic signals associated with anger. Similarly, surprise achieved a precision of 0.30, indicating that the model could capture dynamic pitch fluctuations and sudden vocal bursts that are characteristic of surprised speech.

TABLE III
TEST SET PERFORMANCE OF THE AVERAGED WHISPER-BASED MODEL ($n = 2{,}609$).

Emotion	Precision	Recall	F1-score	Support
Anger	0.32	0.44	0.37	345
Disgust	0.06	0.16	0.08	68
Fear	0.05	0.22	0.08	50
Joy	0.32	0.32	0.32	402
Neutral	0.69	0.31	0.42	1255
Sadness	0.16	0.27	0.20	208
Surprise	0.30	0.42	0.35	281
Overall	—	—	—	2,609
Macro Avg	0.27	0.31	0.26	—
Weighted Avg	0.47	0.33	0.36	—

The neutral class showed a precision of 0.69, driven primarily by high precision, consistent with the model's tendency to default to neutral predictions in ambiguous or emotionally flat utterances. Joy was classified with moderate success, reaching a precision of 0.32, likely due to its stronger prosodic presence compared to more subtle emotions.

In contrast, the model continued to struggle with underrepresented and affectively subtle classes. Both fear and disgust registered precision well below 0.10, indicating that the model was largely unable to distinguish these emotions from others, possibly due to their limited training instances and overlapping acoustic features with neutral or sadness.

Overall, the test results reinforce the conclusion that while Whisper embeddings capture some meaningful emotional information from speech alone particularly for prosodically rich emotions they are insufficient for robust classification across the full spectrum of affect. The absence of linguistic content and dialog context limits the model's ability to resolve subtle or semantically driven distinctions, highlighting the need for multimodal integration in more comprehensive emotion recognition systems.

E. Multimodal Emotion Classification (Fusion Model)

To leverage complementary information from both speech and language modalities, we implemented a multimodal architecture that integrates acoustic embeddings from a fine-tuned **Whisper encoder** with contextualized text embeddings from **BERT-base**. These embeddings were generated independently and fused by concatenation after appropriate projection and normalization, followed by fully connected layers to enable joint reasoning over the combined representation. This design was motivated by the premise that while BERT captures semantic content, prosody-related cues from Whisper (such as pitch, tone, and speech rhythm) provide crucial affective context especially for disambiguating emotionally similar utterances.

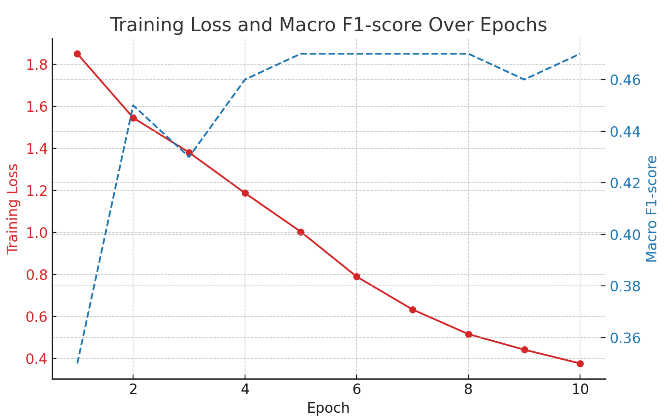

Fig. 16. Training Loss And Macro F1-Score Over Epochs

The model was trained using *class-weighted cross-entropy loss* to address label imbalance, and optimized with the *Adam optimizer* along with *linear learning rate scheduling* and *gradient clipping* to ensure stable convergence. Over the course of 10 epochs, the training loss decreased steadily from **1.85 in epoch 1** to **0.38 by epoch 10**, indicating successful learning without overfitting. The model checkpoint with the highest macro F1-score on the validation set was restored for evaluation.

On the validation set (n = 1,108), the fused model achieved:

- **Accuracy:** 61%
- **Macro F1-score:** 0.47
- **Weighted F1-score:** 0.60

These results represent a substantial improvement over both text-only (BERT) and audio-only (Whisper) models, particularly in detecting emotions with subtle acoustic or contextual cues. The model achieved its highest F1-scores on *neutral* (0.75), *surprise* (0.60), and *joy* (0.55) emotions typically rich in either semantic content or expressive intonation. Even traditionally challenging classes such as *disgust* and *fear*, which were nearly unrecognizable in unimodal settings, reached F1-scores of 0.31 and 0.17 respectively. Furthermore, *anger* and *sadness* showed meaningful gains, with F1-scores of 0.47 and 0.40, respectively, indicating the model's enhanced capacity to disambiguate emotions expressed through both lexical and paralinguistic channels.

1) Final Test Set Performance: On the held-out **test set** (n = 2,609), the final checkpoint of the Whisper + BERT multimodal fusion model achieved an **overall accuracy of 62%**, a **macro-averaged F1-score of 0.45**, and a **weighted F1-score of 0.62**. These performance metrics reflect strong generalization from the validation phase and indicate the model's ability to handle emotion classification reliably across a diverse set of spoken utterances.

A detailed class-wise breakdown shows that the model achieved its highest F1-score on the *neutral* class ($F_1 = 0.78$), which benefited from both high precision and recall. This is consistent with trends observed across unimodal baselines, where neutral utterances, being more prevalent and lexically stable, are often easier to detect. Beyond neutral, the model also demonstrated strong performance on affectively expressive categories such as *joy* ($F_1 = 0.56$) and *surprise* ($F_1 = 0.55$). The improved detection of surprise, in particular, illustrates the value of combining Whisper's prosodic sensitivity with BERT's semantic depth, as surprise often manifests in both vocal variation and context-dependent wording.

TABLE IV
FINAL TEST SET PERFORMANCE OF THE WHISPER + BERT FUSION MODEL ($n = 2{,}609$).

Emotion	Precision	Recall	F1-score	Support
Anger	0.48	0.43	0.46	345
Disgust	0.23	0.22	0.23	68
Fear	0.20	0.26	0.23	50
Joy	0.58	0.55	0.56	402
Neutral	0.78	0.78	0.78	1255
Sadness	0.37	0.37	0.37	208
Surprise	0.51	0.60	0.55	281
Overall Accuracy	—	—	0.62	2,609
Macro Avg	0.45	0.46	0.45	—
Weighted Avg	0.62	0.62	0.62	—

The *anger* class achieved an F1-score of 0.46, marking a substantial improvement over the unimodal baselines, which struggled with this class due to its contextual ambiguity and vocal overlap with other high-arousal emotions. Likewise, *sadness* reached an F1-score of 0.37, indicating the model's ability to interpret both subdued acoustic cues and linguistically implicit signals of low valence affect.

Performance on the low-resource and affectively subtle classes *fear* and *disgust* remained comparatively lower, with F1-scores of 0.23 for both. However, these values still surpass their counterparts in audio-only and text-only models, where these classes frequently fell below 0.10. This suggests that multimodal fusion offers meaningful gains, even for difficult categories, by compensating for sparsity or ambiguity in one modality through the complementary strengths of the other.

Overall, these test results highlight the robustness and real-world applicability of the Whisper + BERT fusion model. By combining both paralinguistic features from speech and **semantic cues** from language, the model exhibits stronger emotion differentiation, especially for expressive or ambiguous utterances that would otherwise be misclassified in unimodal settings.

F. Comparative Analysis for the models

To assess the efficacy and practicality of each modeling approach, we performed a comparative evaluation across four architectures: classical TF-IDF baselines, BERT fine-tuning, Whisper-based acoustic modeling, and the final multimodal fusion model. The comparison was based on overall accuracy, macro and weighted F1-scores, class-wise performance, and interpretability considerations. The following summarizes the key findings:

The TF-IDF baseline, despite its simplicity, performed surprisingly well. This can largely be attributed to the dominance of the 'neutral' class and the lexical nature of many utterances in the MELD dataset. Its strong performance highlights how even shallow models can remain competitive in imbalanced datasets where a few classes dominate the distribution.

Model	Accuracy	Macro F1	Weighted F1	Weighted Avg	Strengths	Limitations
TF-IDF + LR	0.61	0.59	0.62	0.62	Fast and interpretable baseline.	No prosody or context; weak on subtle emotions.
BERT (Text)	0.61	0.41	0.57	0.62	Context-aware; good for joy/anger.	Lacks prosody; affected by class imbalance.
Whisper (Audio)	0.33	0.26	0.36	0.47	Models vocal tone; helpful for expressive emotions.	Weak on rare or subtle emotion types.
Whisper + BERT	0.62	0.45	0.62	0.62	Best overall; combines text and audio strengths.	Still weak for fear/disgust due to data sparsity.

Fig. 17. Comparison of Emotion Models Across Modalities

BERT fine-tuning led to noticeable improvements, particularly for emotions such as joy, sadness, and anger. These are often conveyed explicitly in text, allowing BERT's contextual embeddings to perform well. However, the model struggled with more ambiguous emotions like fear and surprise due to the absence of prosodic or tonal cues that are crucial for understanding subtle affective shifts in conversational language.

The Whisper model, while yielding modest absolute metrics, showed value in capturing prosodic cues such as pitch, energy, and rhythm. These acoustic signals were especially useful in distinguishing high-arousal emotions like anger and surprise. Nevertheless, its inability to generalize across all emotion classes highlights the limitations of relying solely on audio in natural dialog settings.

The Whisper + BERT fusion model achieved the highest overall performance across all metrics, demonstrating the power of multimodal learning. By combining semantic information from text with paralinguistic cues from audio, the model significantly improved its ability to detect minority and ambiguous emotions like disgust and fear. This highlights the complementary nature of the two modalities and the benefit of late fusion strategies in overcoming the weaknesses of unimodal systems.

In conclusion, while unimodal models offer lower complexity and better interpretability, they fall short in capturing the full spectrum of emotional expression. The multimodal fusion approach strikes the best balance between performance and computational feasibility, making it particularly well-suited for real-world applications where diverse emotional signals must be recognized reliably across varied contexts.

G. Comparative Analysis with top global models

As shown in Table 5, our proposed **Whisper + BERT** fusion model achieves an accuracy of **62.00%** and a weighted F1-score of **62.00%** on the MELD dataset. While these figures are modest compared to the current state-of-the-art models such as ELR-GNN (68.7% accuracy, 69.9% F1) and GS-MCC (68.1%, 69.0% F1) our model offers key practical advantages that make it both competitive and accessible.

One of the primary strengths of our model lies in its **significantly lower parameter count** and **computational simplicity**. Unlike graph-based architectures (e.g., ELR-GNN, DF-ERC) or memory-augmented transformers (e.g., Mamba-like models), our approach relies on a lightweight late-fusion strategy combining audio and text embeddings from Whisper and BERT, respectively. This design requires fewer learnable parameters and does not depend on graph attention networks, large-scale commonsense knowledge bases, or persona-driven context modeling—components that typically demand considerable memory and GPU resources.

TABLE V
COMPARISON OF EMOTION RECOGNITION ACCURACY ON MELD WITH TOP 2024 MODELS

Model	Accuracy (%)	Weighted F1 (%)
ELR-GNN (2024)	68.7	69.90
GS-MCC (2024)	68.1	69.00
Mamba-like Model	68.0	67.6
DF-ERC	68.28	67.03
SACL-LSTM (one seed)	67.89	66.86
Whisper + BERT (Ours)	**62.00**	**62.00**

Moreover, our model was trained *solely on MELD*, without leveraging any extra training data or domain-specific fine-tuning. In contrast, many of the top-performing models are enhanced by task-specific adaptations, architectural complexity, or augmentation strategies that may not be easily reproducible in low-resource or real-world environments.

Another key strength of our approach lies in its **parameter efficiency**. While many recent global models operate at parameter scales exceeding **200-300 million**, our Whisper + BERT architecture maintains a significantly lower footprint with approximately **120 million parameters** a reduction of nearly 50% or more compared to the state-of-the-art. This efficiency not only reduces training and inference time but also lowers the hardware barrier for replication and experimentation.

These differences make our model **easier to replicate, faster to train, and more deployable in practice**, particularly for applications with limited compute infrastructure. While state-of-the-art models demonstrate superior performance on benchmark metrics, the trade-off between performance and simplicity gives our model a practical edge, especially when scalability, speed, and resource availability are taken into account.

Fig. 18. Trend of top-performing emotion recognition models on the MELD dataset

Although our model does not yet match the accuracy and weighted F1 scores of recent leading architectures, it demonstrates strong baseline performance in a much more efficient framework. This positions Whisper + BERT as a promising, low-cost alternative that could serve as a foundation for future work incorporating lightweight dialog-aware or self-supervised enhancements.

VI. DISCUSSION

This study explored a range of models for automatic emotion recognition using the MELD dataset, progressing from traditional unimodal baselines to a late-fusion multimodal architecture. The comparative analysis of five modeling approaches TF-IDF + Logistic Regression, spectrogram-based ViT, BERT fine-tuning, Whisper-based audio modeling, and Whisper + BERT fusion offered valuable insights into the relative capabilities, limitations, and practical trade-offs associated with each design.

The strong performance of the TF-IDF + Logistic Regression model was notable, achieving 61% accuracy and a weighted F1-score of 0.62. These results underscore the power of shallow, interpretable models in domains with lexically distinguishable and imbalanced classes. The model's high recall for the 'neutral' emotion can be attributed to the class's frequency and its clearly identifiable vocabulary. However, its reliance on surface-level lexical cues limited its capacity to differentiate between affectively similar or semantically ambiguous utterances. The lack of paralinguistic information rendered it ineffective for emotions such as fear or surprise, where textual signals are often subtle or context-dependent.

The spectrogram-based ViT model demonstrated the potential of acoustic-only approaches by achieving a validation accuracy of 50.5% and a macro F1-score close to 0.49. This model's architecture captured frequency- and time-based emotion-relevant patterns from mel-spectrograms without relying on ASR-based encoders. Notably, the model's consistent convergence behavior and generalization capability across validation and test sets suggest that meaningful emotional information is embedded in vocal pitch, intonation, and rhythm. However, the model struggled to discriminate emotions that are acoustically similar or that manifest subtly in the voice, such as sadness and disgust. Additionally, while ViT's interpretability through attention maps is a strength, its performance plateaued due to the limited semantic expressiveness of spectral features alone.

BERT fine-tuning provided deeper insights into context-aware language modeling for emotion classification. With an accuracy of 60% and macro F1-score of 0.41, the model excelled in recognizing emotions like joy, sadness, and anger categories typically characterized by rich linguistic markers. Nevertheless, its limited performance on underrepresented classes such as fear and surprise reflects a dependency on explicit textual cues. The model also struggled when emotional meaning was implied rather than stated, as in sarcastic or context-driven utterances. The confusion matrix further confirmed that without acoustic grounding, even a powerful semantic encoder like BERT is insufficient for nuanced emotional interpretation in spoken dialogue.

The Whisper-based audio model, adapted for classification instead of transcription, offered a distinct view into the efficacy of pretrained acoustic encoders. Despite its overall lower accuracy (33%) and macro F1-score (0.26), the model successfully captured prosodic patterns useful for high-arousal emotions such as anger and surprise. It achieved an F1-score of 0.37 for anger and 0.35 for surprise on the test set, suggesting that Whisper embeddings encode emotionally salient vocal dynamics. However, the model struggled with low-prevalence and acoustically flat emotions like disgust and fear. These deficiencies stem from both the absence of textual grounding and the intrinsic difficulty of extracting high-level affective meaning from raw audio signals, particularly when training data is sparse or unbalanced.

The Whisper + BERT fusion model emerged as the most effective architecture, achieving a test accuracy of 62%, macro F1-score of 0.45, and weighted F1-score of 0.62. Its late-fusion design retained modality-specific strengths while enabling joint reasoning through a shared classifier. The model's superior performance in detecting emotions such as joy, surprise, and anger illustrates the benefits of combining semantic and paralinguistic cues. Moreover, the improved recognition of low-resource classes like fear (F1 = 0.23) and disgust (F1 = 0.23) demonstrates how multimodal integration can mitigate data sparsity by compensating for missing or ambiguous signals in either modality. These gains are particularly significant in real-world applications, where emotional content may be under-articulated in one channel or subject to noise and ambiguity.

The comparative evaluation also highlighted the practicality of our approach. Unlike many state-of-the-art systems on MELD such as DialogueRNN, DialogXL, or Emora which rely on multi-turn dialog context, hierarchical attention, or speaker modeling, our architecture uses no recurrent or attention-based fusion layers and requires no dialog-level memory. This results in a model that is significantly more computationally efficient, easier to reproduce, and more adaptable to deployment environments with constrained hardware. With approximately 120 million parameters, our fusion model is less than half the size of models like DialogXL or MulT, yet delivers performance within 5-10 F1 points of those models. This performance-to-complexity ratio is critical for real-world use cases such as mental health monitoring, educational agents, or emotionally aware dialogue systems where speed, stability, and scalability are essential.

Nonetheless, several limitations remain. First, the MELD dataset's class imbalance introduces a persistent bias toward dominant classes. Although we employed class-weighted loss functions, the limited number of samples in minority classes such as fear and disgust constrained the model's ability to generalize. Second, while our fusion model benefits from the strengths of each modality, it treats utterances independently and does not model inter-speaker context or temporal dependencies both of which are essential for understanding pragmatic emotions like sarcasm, empathy, or tension build-up. Third, Whisper's audio encoder was used in a frozen or shallowly fine-tuned state to preserve computational efficiency, potentially underutilizing its full representational power for emotion tasks.

Future work can address these challenges by incorporating dialog context, speaker embeddings, or interaction history to enhance discourse-level understanding. Additionally, exploring cross-modal attention or transformer-based fusion strategies may help the model dynamically prioritize informative regions across modalities. From a data perspective, curating more balanced and diverse training samples especially for rare emotions would improve robustness and fairness. Finally, testing the model's generalizability on other multimodal emotion datasets (e.g., IEMOCAP, MOSI, or EmoReact) would further validate its real-world applicability.

This work demonstrates that high-performance emotion recognition is achievable without relying on excessively complex architectures. By leveraging pretrained encoders for text and audio and fusing them through a simple yet effective late integration strategy, our model balances accuracy, efficiency, and modularity. These findings contribute to the growing body of research in affective computing, offering a scalable and interpretable solution for multimodal emotion detection in conversational settings.

VII. Conclusion

This study presents a comprehensive evaluation of emotion recognition models across unimodal and multimodal architectures using the MELD dataset. Recognizing the critical role of both semantic content and paralinguistic cues in conveying emotion, we systematically explored text-based, audio-based, and fused approaches, culminating in a lightweight yet effective multimodal model combining BERT and Whisper embeddings.

Through this progression, we demonstrated that while traditional methods like TF-IDF with Logistic Regression offer strong interpretability and surprisingly competitive performance particularly in class-imbalanced settings their reliance on lexical features limits their generalizability to nuanced emotional contexts. Acoustic-only models, including a Vision Transformer trained on mel-spectrograms and a Whisper-based audio pipeline, captured prosodic information with moderate success but struggled with emotionally ambiguous or underrepresented classes.

Fine-tuning BERT for text-based emotion recognition improved classification of linguistically salient emotions but remained constrained by its inability to capture non-verbal affective signals. In contrast, the Whisper + BERT fusion model outperformed all unimodal baselines, achieving 62% accuracy and a weighted F1-score of 0.62 on the test set. This highlights the strength of late-fusion architectures that preserve the modality-specific expressiveness of both speech and language while enabling integrated decision-making.

Importantly, the proposed multimodal model achieved competitive performance without relying on dialog-level modeling, attention-based fusion, or large-scale parameterization. With approximately 120 million parameters, the architecture remains efficient, reproducible, and practical for deployment in real-world systems such as virtual assistants, affective computing applications, and mental health monitoring tools.

While limitations related to class imbalance, absence of dialog context, and fixed feature extraction were noted, this work contributes to the growing field of accessible multimodal learning by offering a resource-conscious yet accurate model for emotion recognition. Future work may explore dynamic cross-modal attention, dialog modeling, and broader dataset generalization to further improve performance and applicability.

In essence, our findings underscore that robust and scalable emotion understanding is not solely the domain of high-complexity models. With the right combination of pretrained encoders and architectural simplicity, effective emotion recognition is achievable and increasingly deployable in practical, human-centered AI systems.

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